BEFORE THE WORKSHOP / INSTALLATION GUIDE

Hyperledger Fabric

Get your laptop ready.

Follow these steps in order before workshop day. By the end, your laptop will run a small Fabric network and read a record from its ledger.

Who this guide is for
Windows 11 laptops with an Intel or AMD 64-bit processor.
If you use Windows on ARM, macOS, Linux, or a college-managed laptop, ask the instructor for the matching setup before starting.

Before you begin

- **Memory:** 8 GB minimum; 16 GB recommended for a smoother workshop.
- **Storage:** keep at least 25 GB free as a workshop allowance.
- **Access:** administrator permission, a reliable internet connection, and your laptop charger.
- **Time:** allow 60–90 minutes, plus download time. Finish at least one day before the workshop.

Your setup route

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Workshop setup Ubuntu 24.04 LTS on WSL 2; Node.js 22.23.3; Fabric 2.5.16; Fabric CA 1.5.17. Use a current supported Docker Desktop with the WSL 2 backend.

What this prepares The workshop will build a demo credits-transfer project. The asset example in this guide is only an installation check. No real money, wallet, or paid blockchain account is needed.

Official help: Docker system requirements. Guide edition: 17 October 2026.

01 / WINDOWS SETUP

Install Ubuntu with WSL 2

WSL lets Ubuntu run inside Windows. You do not need to dual boot your laptop.

1 Open PowerShell as administrator

Open the Windows Start menu, search for **PowerShell**, right-click it, and choose **Run as administrator**. Approve the Windows prompt.

RUN IN: WINDOWS POWERSHELL – ADMINISTRATOR

```
wsl --install -d Ubuntu-24.04
```

Restart Windows if requested. After restarting, open **Ubuntu 24.04 LTS** from the Start menu to finish its first launch.

2 Create your Ubuntu login

Choose a simple lowercase username, then create a password and enter it again. This is your Ubuntu login; it can differ from your Windows login. **Password tip** Nothing appears while you type the Ubuntu password, including no dots. Type it normally and press Enter. Remember it for commands starting with `sudo`.

3 Confirm that Ubuntu uses version 2

Return to Windows PowerShell and run:

RUN IN: WINDOWS POWERSHELL

```
wsl --update
wsl --list --verbose
```

Check before continuing
The list contains `Ubuntu-24.04` and its `VERSION` column says `2`. The state may be `Running` or `Stopped`; both are fine.

If it says version 1 Run the following in PowerShell, then check the list again. If your distribution has another name, use its exact name from the list.

```
wsl --set-version Ubuntu-24.04 2
```

4 Open your Ubuntu terminal

Launch **Ubuntu 24.04 LTS** from Start. You should see a prompt resembling `yourname@laptop: $`. From page 4 onward, all commands run here. **Already have Ubuntu?** Check it with `lsb_release -ds` inside Ubuntu. If it is Ubuntu 24.04 on WSL 2, continue. Keep existing distributions and files; do not uninstall or reset them.

Official help: Microsoft: Install WSL. If installation fails, see page 8.

02 / CONTAINERS

Set up Docker Desktop

Docker runs the Fabric peers and ordering service in containers.

5 Install Docker Desktop on Windows

1. Visit Docker Desktop for Windows and download the **x86_64** installer.
2. Run the installer. Keep **Use WSL 2 instead of Hyper-V** selected if the option appears.
3. Complete installation and restart Windows if requested.
4. Open **Docker Desktop**, complete the welcome screens, and wait until its engine is running.

6 Enable Ubuntu integration

In Docker Desktop, open **Settings**:

1. Under **General**, enable **Use the WSL 2 based engine** if shown.
2. Under **Resources** → **WSL Integration**, enable integration for **Ubuntu-24.04**.
3. Select **Apply & restart** if shown. Close and reopen your Ubuntu terminal.

If WSL Integration is missing Docker may be using Windows containers. Open the Docker tray menu and switch to **Linux containers**, then check Settings again. Menu wording can vary by release.

7 Test Docker from Ubuntu

RUN IN: UBUNTU

```
docker --version
docker compose version
docker info --format '{{.OSType}}'
docker run --rm hello-world
```

Check before continuing
Both version commands print a version. The operating-system check prints `linux`. The final command prints `Hello from Docker!`. Its first run downloads a small image.

Use one Docker installation Docker Desktop supplies the engine for Ubuntu. Do not also install `docker.io`, Docker Engine, or a second Docker Compose inside Ubuntu for this guide. Run workshop Docker commands without `sudo`.

Official help: Docker: WSL 2 backend / Fabric: Windows prerequisites.

03 / DEVELOPER TOOLS

Prepare Ubuntu

Install these tools inside Ubuntu, even if Node.js or Git already exists on Windows.

8 Install the basic tools

RUN IN: UBUNTU

```
sudo apt update
sudo apt install -y git curl jq build-essential ca-certificates
```

Enter your Ubuntu password if asked. Wait until the command finishes and the prompt returns.

9 Install Node.js through nvm

Node.js runs JavaScript tools and the application that connects to Fabric. `nvm` keeps its installation inside your Ubuntu account.

RUN IN: UBUNTU

```
curl -fsSL \
  https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.8/install.sh \
  -o /tmp/nvm-install.sh
bash /tmp/nvm-install.sh
source ~/.bashrc
nvm install 22.23.3
nvm alias default 22.23.3
nvm use 22.23.3
```

How to copy commands Copy one complete block at a time. A backslash (\) at the end of a line means the command continues on the next line. Do not add anything after it.

10 Check the tools

RUN IN: UBUNTU

```
git --version
curl --version
jq --version
node --version
npm --version
command -v node
```

Check before continuing
Node prints `v22.23.3`; npm prints a version. The Node path starts with `/home/` and includes `.nvm/`. It should not point to a Windows folder under `/mnt/c/`.

Keep your project in Ubuntu Use the home folder `~/fabric-workshop` in the next step. Avoid OneDrive and Windows-mounted project folders for this workshop.

Recommended editor Install Visual Studio Code for Windows. In VS Code, open Extensions (`Ctrl+Shift+X`), search for **WSL**, and install the extension published by **Microsoft**. It lets you edit your Ubuntu project from Windows.

Official help: nvm installation instructions / Node.js 22.23.3 release.

04 / FABRIC TOOLS

Download Hyperledger Fabric

Keep Docker Desktop running. This stage downloads several large files.

11 Get the workshop sample revision

This fixed revision keeps everyone on the same sample scripts. Git may say `detached HEAD` after checkout; that is expected.

RUN IN: UBUNTU

```
mkdir -p ~/fabric-workshop
cd ~/fabric-workshop
git clone https://github.com/hyperledger/fabric-samples.git
cd fabric-samples
git checkout 3c63eac4e303e8348c42637d7ed337727645b9ce
```

Folder already exists? Do not delete it. If it belongs to another project, use a new parent folder and ask the instructor before changing its revision. The remaining commands assume the path shown above.

12 Download the official installer

Stay in `~/fabric-workshop/fabric-samples`. The command saves the installer one folder above it.

RUN IN: UBUNTU

```
curl -fsSLo ../install-fabric.sh \
  https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
bash ../install-fabric.sh --fabric-version 2.5.16 \
  --ca-version 1.5.17 binary docker
```

The installer downloads Fabric command-line tools, configuration files, and container images. Wait for it to finish. Some images download from GitHub Container Registry.

13 Confirm the downloads

RUN IN: UBUNTU

```
./bin/peer version
./bin/fabric-ca-client version
git rev-parse --short=12 HEAD
docker image inspect hyperledger/fabric-peer:2.5.16 \
  --format '{{.RepoTags}}'
docker image inspect hyperledger/fabric-ca:1.5.17 \
  --format '{{.RepoTags}}'
```

Check before continuing
Peer reports `2.5.16`, CA client reports `1.5.17`, and the revision starts with `3c63eac4e303`. Both image checks print tags without an error.

Use the exact versions Do not replace these numbers with `latest`, or switch the samples to another branch. Java and Go are not required for our JavaScript project.

Official help: Fabric installation / Pinned sample source.

05 / FIRST NETWORK

Start your test network

This is a local learning network on your laptop. It is not a production deployment.

14 Create the network and channel

RUN IN: UBUNTU

```
cd ~/fabric-workshop/fabric-samples/test-network
./network.sh up createChannel -ca -c mychannel \
  -i 2.5.16 -cai 1.5.17
docker ps --format 'table {{.Names}}\t{{.Status}}'
```

Check before continuing
The script finishes successfully and joins `mychannel`. The running-container list includes `peer0.org1.example.com`, `peer0.org2.example.com`, and `orderer.example.com`, plus the certificate-authority containers.

15 Deploy the JavaScript sample contract

Stay in the same `test-network` folder. This uses the official asset example to confirm that JavaScript chaincode can be installed and started.

RUN IN: UBUNTU

```
./network.sh deployCC -c mychannel -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccl javascript
```

The first deployment may take several minutes while npm packages and a chaincode runtime image download. Leave the terminal open until it returns to the prompt.

Check before continuing
The output reports a committed chaincode definition for `basic`, version `1.0`, sequence `1`, on `mychannel`. There should be no final installation or commit failure.

What is running?

Two peer organizations	Store the ledger and execute the agreed contract.
Ordering service	Places transactions into an agreed sequence.
Certificate authorities	Issue identities for this test network.
`mychannel`	The shared ledger used for this check.
`basic`	The sample smart contract you just deployed.

If a step fails Stop at that step and use page 8. Do not continue to the next command assuming that deployment worked.

Official help: Fabric test-network tutorial. Use the ordinary Raft setup shown here, without `-bft`.

06 / READINESS TEST

Write once. Read it back.

These commands use the sample identities created on page 6. Keep this Ubuntu terminal open.

16 Load the test identity and tools

RUN IN: UBUNTU

```
cd ~/fabric-workshop/fabric-samples/test-network
export PATH="$PWD/../bin:$PATH"
export FABRIC_CFG_PATH="$PWD/../config/"
source scripts/envVar.sh
setGlobals 1
```

17 Create the example records

Copy this entire command, including every continued line:

RUN IN: UBUNTU

```
peer chaincode invoke -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls --cafile "$ORDERER_CA" \
  -C mychannel -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles "$PEER0_ORG1_CA" \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles "$PEER0_ORG2_CA" \
  -c '{"function":"InitLedger","Args":[]}' --waitForEvent
```

Check before continuing
The command reports a successful invocation with status 200 and a committed transaction with status `VALID`. If it times out, see page 8 before repeating.

18 Read a record

RUN IN: UBUNTU

```
peer chaincode query -C mychannel -n basic \
  -c '{"function":"ReadAsset","Args":["asset1"]}'
```

Check before continuing
A JSON record appears with `ID` equal to `asset1` and `Owner` equal to `Tomoko`. Field order may vary. Save a screenshot of this result for the readiness check. Your installation is ready.

19 Clean up the practice network

After saving your screenshot, run this from `test-network`:

```
./network.sh down
```

This resets the practice ledger It deletes the sample network and its test identities. Keep the downloads and Docker images. Bring your readiness screenshot, laptop, and charger to the workshop.

Official help: Sample identity helper / Sample contract.

HELP / QUICK FIXES

If something gets stuck

Keep the full error message. Fix the failed stage before continuing.

WSL reports virtualization or error `0x80370102`

Restart Windows. In Task Manager → Performance → CPU, check that virtualization is enabled. If it is disabled, ask the instructor or campus IT to help enable it in the laptop firmware. Then follow Microsoft's WSL troubleshooting steps.

Ubuntu does not appear in Docker's WSL Integration

Open Ubuntu once to finish its setup. In PowerShell, run `wsl -list -verbose` and confirm version 2. Check that Docker Desktop uses Linux containers, then enable that Ubuntu distribution in Settings.

`docker: command not found` or cannot connect to Docker

Open Docker Desktop and wait for its engine. Recheck Ubuntu integration, then close and reopen Ubuntu. Run the checks on page 3 again. Do not fix this by installing another Docker engine inside Ubuntu.

`nvm: command not found` or the wrong Node version

In Ubuntu, run `source ~/.bashrc`, then `nvm use 22.23.3`. If it still fails, repeat step 9. Use `command -v node` to confirm the Ubuntu installation.

A download fails or returns a network / certificate error

Check internet access and free storage. Retry the failed download after connectivity returns. Campus Wi-Fi may block GitHub, container registries, or npm; ask IT for access or use an approved alternative connection. Keep certificate verification enabled.

Fabric says a port is in use or an old network exists

Ask the instructor if other Fabric work is already running. For this disposable practice network only, run `./network.sh down` from `test-network`, then repeat steps 14–18. This deletes the practice ledger.

Transaction times out, or the asset cannot be found

Check `docker ps` and confirm the peers and orderer are running. Reload step 16 if you opened a new terminal. Retry the read after a few seconds. If records are still missing, share the invocation output from step 17 with the instructor before resetting anything.

Ask for help with these details
Your Windows version, the step number, and a screenshot of the full error. Include `wsl -list -verbose` from PowerShell and `docker -version`, `docker compose version`, and `node -version` from Ubuntu. Keep passwords and private-key files out of screenshots.

Official references

Microsoft WSL / Docker Desktop / Docker WSL integration

Fabric prerequisites / Fabric installer / Fabric test network